

QLog

draft-ietf-quic-qlog-main-schema-14

draft-ietf-quic-qlog-quic-events-13

draft-ietf-quic-qlog-h3-events-13

QUIC – IETF 126 – 2026-07

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Since IETF 125

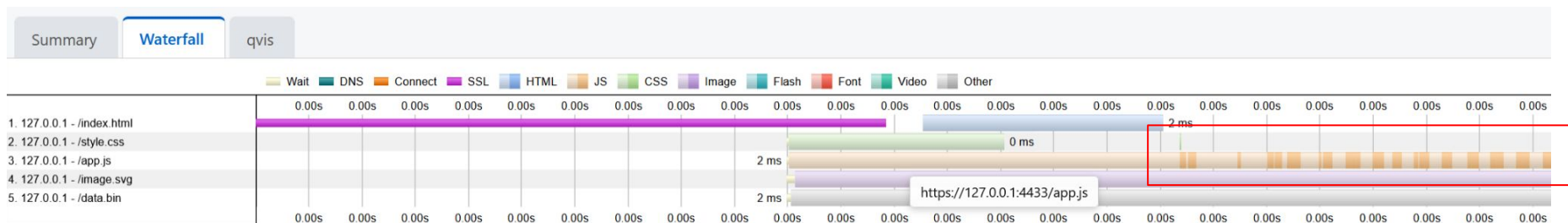
- Published new drafts
- Key changes
 - Numerous design changes to individual events
 - Numerous editorial improvements
- Key non-changes
 - Foundations and design - no major changes are planned
 - JSON-SEQ vs. JSON issue closed with no action

Implementations/Interop

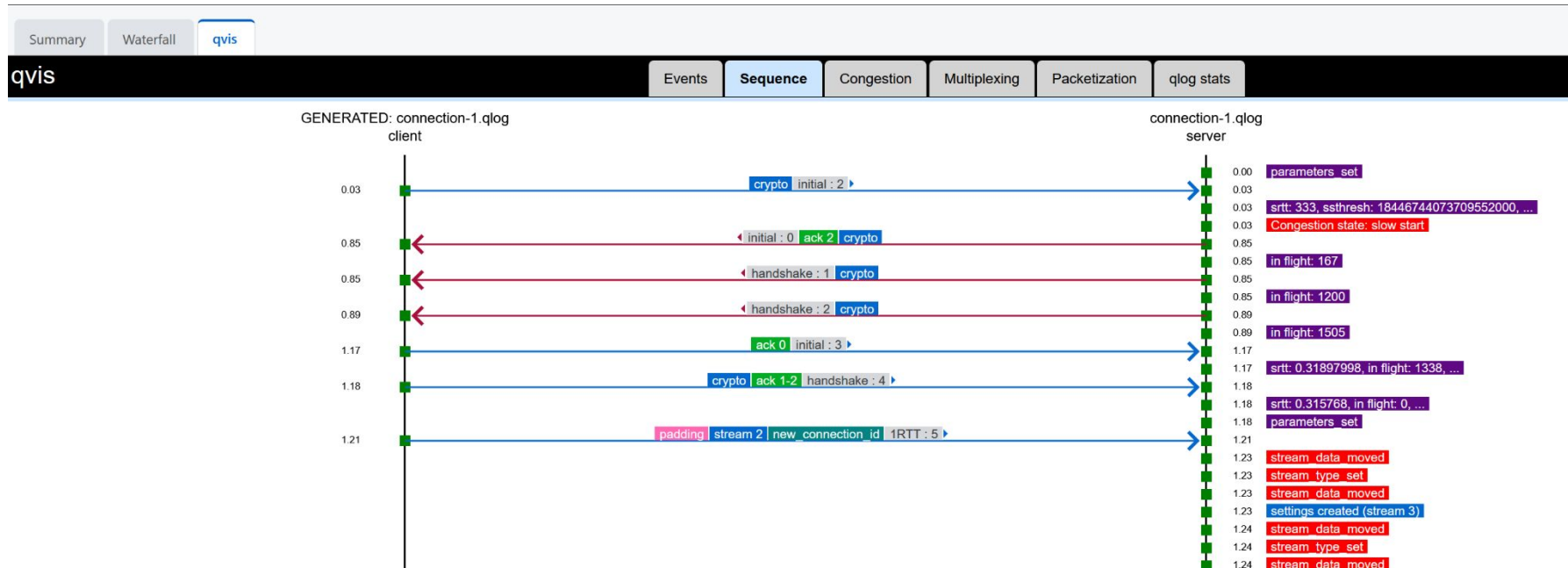
- Varying support for qlog versions
 - Now is a really good time to update to latest
 - AI / LLM does a very good job at handling these structured definitions
- New parser tool
 - Pat Meenan - Waterfall tools - <https://github.com/pmeenan/waterfall-tools>
 - Please test qlog features and provide feedback on <https://github.com/pmeenan/waterfall-tools/issues/12>

Waterfall tools example

- [Live link](#)



Waterfall tools X QVis



Waterfall tools X netlog viewer

SummaryWaterfallqvis

qvisEventsSequenceCongestionMultiplexingPacketizationqlog stats

Filter events (case-insensitive substring of type + summary)All categories

#	time (ms)	event type	summary
0	0.000	quic:parameters_set	initiator=local max_idle_timeout=30000 initial_max_data=10000000 initial_max_streams_bidi=...
1	0.028	quic:packet_received	initial #2 · CRYPTO
2	0.028	quic:recovery_metrics_updated	min_rtt=333 smoothed_rtt=333 rtt_variance=166.5 pto_count=0 congestion_window=13500 sst...
3	0.028	quic:congestion_state_updated	new=slow_start
4	0.851	quic:packet_sent	initial #0 · ACK, CRYPTO
5	0.851	quic:recovery_metrics_updated	bytes_in_flight=167
6	0.851	quic:packet_sent	handshake #1 · CRYPTO
7	0.851	quic:recovery_metrics_updated	bytes_in_flight=1200
8	0.886	quic:packet_sent	handshake #2 · CRYPTO
9	0.886	quic:recovery_metrics_updated	bytes_in_flight=1505
10	1.170	quic:packet_received	initial #3 · ACK
11	1.170	quic:recovery_metrics_updated	min_rtt=0.31897998 smoothed_rtt=0.31897998 latest_rtt=0.31897998 rtt_variance=0.1594899...
12	1.179	quic:packet_received	handshake #4 · ACK, CRYPTO
13	1.179	quic:recovery_metrics_updated	min_rtt=0.293288 smoothed_rtt=0.315768 latest_rtt=0.293288 rtt_variance=0.12604 bytes_in_...
14	1.179	quic:parameters_set	initiator=remote cipher=AES128_GCM max_idle_timeout=30000 initial_max_data=10000000 i...
15	1.214	quic:packet_received	1RTT #5 · NEW_CONNECTION_ID, STREAM(2, 19B), PADDING
16	1.226	quic:stream_data_moved	stream 3 · off 0 · 1B · application→transport
17	1.228	http3:stream_type_set	initiator=local stream_id=3 stream_type=control
18	1.233	quic:stream_data_moved	stream 3 · off 1 · 18B · application→transport
19	1.234	http3:frame_created	stream 3 SETTINGS

#0 · +0.000 ms · quic:parameters_set

```
{
  "time": 0,
  "name": "quic:parameters_set",
  "data": {
    "initiator": "local",
    "original_destination_connection_id": "7583215733fc13038ece39565bb2cf6b",
    "disable_active_migration": true,
    "max_idle_timeout": 30000,
    "max_udp_payload_size": 1350,
    "ack_delay_exponent": 3,
    "max_ack_delay": 25,
    "active_connection_id_limit": 2,
    "initial_max_data": 10000000,
    "initial_max_stream_data_bidi_local": 1000000,
    "initial_max_stream_data_bidi_remote": 1000000,
    "initial_max_stream_data_uni": 1000000,
    "initial_max_streams_bidi": 100,
    "initial_max_streams_uni": 100,
    "unknown_parameters": []
  },
  "timeText": "0.00"
}
```

This event is not associated with a stream

Open issues

Only 4 open issues:

- 0-RTT is a bit ambiguous in -01 [#27](#)
- Allow the logging of Session Ticket contents and stored transport params/SETTINGS [#121](#)
- consider moving tuple information to QUICPacketSent / QUICPacketReceived / QUICPacketDropped event [#501](#)
- Make frames_processed consistent across QUIC and HTTP/3 [#517](#)

Frames_processed consistency issue [#517](#)

- History:
 - QUIC Frame traditionally logging in packet_sent / packet_received events as an array
 - Trying to use qlog for qmux => no packets => issue [#507](#)
 - Existing frames_processed event - can we repurpose to signal sent or received?
- Now:
 - QUIC: packet_sent, packet_received, frames_created, frames_parsed, frames_processed
 - HTTP/3: frame_created, frame_parsed

Frames_processed consistency issue [#517](#)

- QUIC frames_processed

The `frames_processed` event can be used to signal internal state change not resulting directly from the actual parsing of a frame (e.g., the frame could have been parsed, data put into a buffer, then later processed and logged). Using `frames_processed` in addition to `packet_received` and/or `frames_parsed` allows an implementation to log its full sequence of events.

Frames_processed consistency issue [#517](#)

Proposals:

1. Remove QUIC frames_processed
2. Add HTTP/3 frames_processed
3. Do nothing

Questions / Next Steps

Please help close the remaining issues

WGLC

Support new and emerging qlog drafts in the wider IETF:

- [draft-ietf-tsvwg-careful-resume-qlog](#)
- [draft-kuehlewind-happy-qlog](#)
- [draft-engelbart-qlog-roq-events-00](#)
- [draft-pardue-moq-qlog-moq-events](#)
- [draft-pardue-httpbis-qlog-h2-events](#)